

Optical-Analog-Optical Wavelength Converter for Multi-Domain Optical Direct-Connect Boundaries: 100G/400G Multi-FEC Characterization and Network Demonstration

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Abstract *We experimentally evaluate an Optical-Analog-Optical wavelength converter as a boundary node for multi-domain optical direct-connect networks. 100G/400G multi-FEC and multi-vendor measurements indicate OSNR penalty variations correlate more strongly with transceiver-dependent characteristics than with FEC scheme. Multi-domain network demonstration confirms measurable margin and frequency-offset detection capability. ©2026 The Author(s)*

Introduction

The IOWN All-Photonics Network (APN) aims to realize end-to-end optical direct-connect services across independently operated network domains [1]. By reducing optical-to-electrical conversion at domain boundaries, this architecture has the potential to decrease equipment count, power consumption, cost, and latency. However, removing conventional transponder-based points of interconnection (POIs) also reduces the clear functional separation that previously ensured wavelength isolation and operational autonomy between domains. A boundary device is therefore necessary to maintain domain independence while enabling optical direct-connect paths.

Wavelength conversion is well suited for such boundary devices because it enables wavelength-plan decoupling and independent resource management across domains [2], [3]. In multi-domain environments, boundary nodes must interconnect heterogeneous network segments and interface with transceivers independently selected by each operator. Consequently, signals from different vendors, with various modulation formats, baud rates, and FEC schemes, may coexist at the boundary. In practical deployments, interoperability is often achieved through uniform specifications. However, in multi-domain direct-connect scenarios, such uniformity cannot always be assumed. It is therefore important to understand how the performance impact of wavelength conversion relates to FEC schemes and to transceiver-dependent performance characteristics.

Although OAO (Optical-Analog-Optical) wavelength conversion has been investigated for flexible optical networking, experimental demonstration under multi-vendor and multi-FEC conditions remains limited for 400G-class signals, particularly in realistic multi-domain network configurations. In addition, operational aspects such as

frequency-offset detection at domain boundaries have not been widely reported.

In this paper, we experimentally evaluate an OAO wavelength converter (WC) implemented in a CFP2 form factor as a boundary device for multi-domain optical direct-connect networks. OSNR penalties are characterized using 100G DP-QPSK and 400G DP-16QAM signals under multi-FEC and multi-vendor conditions through ASE-loaded OSNR-sweep measurements. Furthermore, a multi-domain network configuration using commercial ROADMs equipment is constructed to demonstrate end-to-end connectivity and operational robustness, including frequency-offset detection capability.

Experimental Setup

A prototype OAO WC supporting multi-rate operation at 100G and 400G was developed and implemented in a CFP2 compatible form factor. The device enables intra-C-band wavelength conversion over the 191.35–196.10 THz range. It performs optical-to-electrical analog conversion followed by electrical analog-to-optical conversion without digital demodulation, enabling modulation- and FEC-transparent operation.

Three types of experiments were conducted to evaluate the suitability of the OAO WC as a multi-domain boundary node.

A. Pre-FEC BER versus OSNR Characterization
To quantify the performance impact of wavelength conversion, OSNR-sweep measurements were performed using ASE loading as shown in Fig. 1(a). The evaluated signal configurations were as follows:

400G signals (DP-16QAM):

- T1: 400G ZR+ (oFEC, 60.13GBd) [4], 400G ZR (CFEC, 59.84GBd) [5]
- T2: 400G OpenROADM (oFEC, 63.13GBd) [6]

100G signals (DP-QPSK):

- T1: 100G ZR+ (oFEC, 30.07GBd)
- T2: 100G ZR+ (oFEC, 30.07GBd), 100G OpenROADM (SC-FEC [7], 27.95GBd)

Two configurations were compared:

1. Back-to-back (B2B): direct transceiver -to-transceiver connection

2. With WC: transceiver – WC – transceiver

OSNR was varied by adding ASE noise, and Pre-FEC BER was measured at the receiving transceiver.

The OSNR penalty was defined as

$$\Delta\text{OSNR} = \text{OSNR}_{\text{req}}^{\text{WC}} - \text{OSNR}_{\text{req}}^{\text{B2B}} \quad (1)$$

at the defined Pre-FEC BER threshold for each FEC scheme. The Pre-FEC BER thresholds were set to 2.0×10^{-2} for oFEC and 1.0×10^{-2} for CFEC, and to 4.0×10^{-3} for SC-FEC. OSNR was measured at the WC input using an OSA with 0.1-nm resolution bandwidth.

B. Frequency-Offset Detection

To evaluate operational robustness at the domain boundary, the input optical frequency to the WC was swept around the configured center wavelength, as shown in Fig. 1(a). The output response and alarm-detection range were recorded to assess the monitoring capability under frequency misconfiguration conditions.

C. Optical Direct-Connect Paths through Multi-Domain ROADM Network

To emulate a realistic multi-domain environment, a ROADM-based network was constructed, as shown in Fig. 1(b). The testbed consisted of:

- Two domains (Domain A and Domain B)
- Two commercially available ROADMs per domain
- 75 km of SMF per domain (total 150 km), emulating a representative metro-span configuration within the typical 0–200 km reach defined for OpenZR-class interfaces.
- Wavelength conversion deployed at the domain boundary via add/drop ports
- Multi-vendor 400G transceivers were used, representing independent implementations compliant with OpenZR-class specifications.

Pre-FEC BER was measured with and without the WC to evaluate the Pre-FEC BER margin

relative to each FEC threshold and to verify end-to-end connectivity using commercial ROADM equipment.

Results

A. OSNR Penalty Characterization (400G-Centric Analysis)

Figure 2(a)(b) shows the OSNR versus Pre-FEC BER characteristics for the evaluated 400G configurations. The OSNR penalty was extracted at each defined Pre-FEC BER threshold using (1).

For 400G signals, the measured penalties were:

- T1 400G ZR+ (oFEC): $\Delta\text{OSNR} = 1.7\text{ dB}$
- T1 400G ZR (CFEC): $\Delta\text{OSNR} = 3.2\text{ dB}$
- T2 400G OpenROADM (oFEC): $\Delta\text{OSNR} = 2.6\text{ dB}$

Although the penalty values differ among configurations, the OSNR–BER curves show similar slopes for different FEC schemes within the same transceiver. When compared at identical OSNR levels rather than at BER thresholds, the BER degradation introduced by wavelength conversion is nearly identical across FEC schemes within the same transceiver, indicating that the added noise contribution is largely independent of the selected FEC scheme. Instead, the variation in ΔOSNR is primarily associated with transceiver-dependent characteristics.

These results suggest that, under the evaluated conditions, the OSNR penalty variations are more strongly correlated with transceiver-dependent characteristics than with FEC scheme.

B. 100G Reference Evaluation

Figure 2(c) summarizes the corresponding results for 100G signals. The measured OSNR penalties are sufficiently lower than those observed for 400G. Both ZR+ and ZR modules with oFEC, as well as SC-FEC configurations, exhibit stable operation across the evaluated OSNR range. Similar to the 400G results, no clear dependence of the conversion penalty on the FEC scheme is observed within the same transceiver family.

These results confirm consistent multi-rate penalty characteristics across modulation formats and FEC configurations.

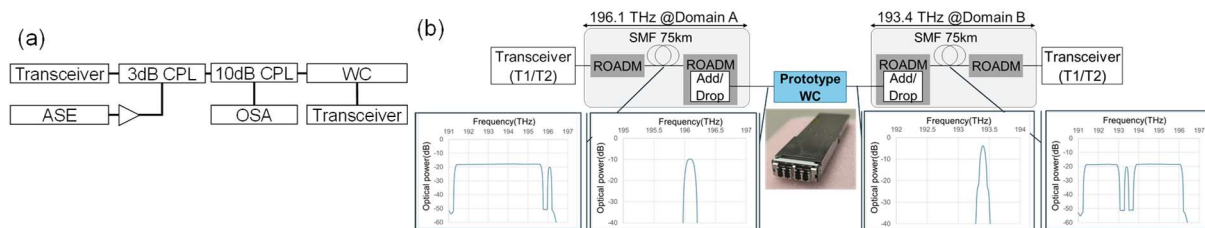


Fig. 1: Evaluation configuration for (a) OSNR Penalty Characterization and Frequency Offset Detection (b) ROADM-based multi-domain demonstration.

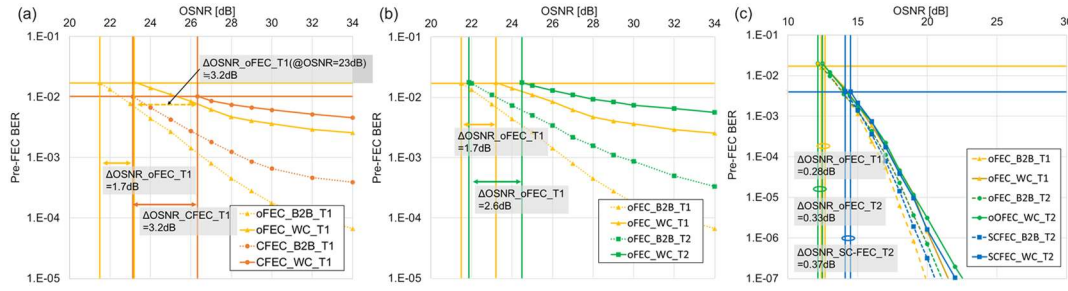


Fig. 2: OSNR versus Pre-FEC BER characteristics for 400G signals (a) FEC comparison (b) Vendor comparison (c) 100G reference.

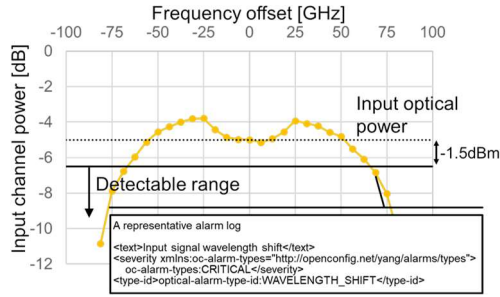


Fig. 3: Input channel power versus frequency offset.

C. Frequency Offset Detection Capability

Figure 3 shows the measured input channel power and a representative alarm log. The alarm threshold was defined relative to the optical input power, and -1.5 dBm was adopted as the input channel power threshold. For example, when the optical input power was -5.0 dBm, the corresponding threshold was -6.5 dBm.

Within a frequency offset of 0 – 50 GHz, the detected channel power remained close to the input optical power. Beyond 50 GHz, the power decreased rapidly, and for offsets exceeding approximately ± 67.5 GHz, it fell below the threshold, triggering the alarm and output shutoff.

These results demonstrate a monitoring capability suitable for domain-boundary deployment, enabling reliable detection of frequency misconfiguration or abnormal signal injection while preventing unintended cross-domain misconnection and preserving operational isolation between domains.

D. Multi-Domain ROADM Demonstration

The multi-domain ROADM demonstration using commercial equipment is illustrated in Fig. 1, and Table 1 summarizes the measured Pre-FEC BER values. The two endpoint transceivers were configured with different line-side wavelengths ($f_A = 196.1$ THz and $f_B = 193.4$ THz), and end-to-end connectivity was achieved by wavelength

translation at the boundary node ($f_A \rightarrow f_B$). In all configurations:

- Pre-FEC BER remained below the corresponding FEC thresholds.
- A measurable margin to the threshold was maintained (0.29 – 0.41 decades in Table 1).
- End-to-end connectivity was successfully established across domains.

Although minor BER degradation was observed when the WC was inserted, the degradation did not compromise FEC operability. These results demonstrate the practical feasibility of OAO-based wavelength conversion in commercial ROADM multi-domain deployments.

Conclusion

We demonstrated an Optical-Analog-Optical wavelength converter in a CFP2-compatible form factor as a boundary node for multi-domain optical direct-connect networks. Multi-rate evaluations using 100G and 400G signals under multi-FEC and multi-vendor conditions showed that variations in the observed OSNR penalty are more strongly associated with transceiver-dependent characteristics than with the selected FEC scheme.

An experimental demonstration in a multi-domain network configuration using commercial ROADM equipment confirmed stable end-to-end connectivity and measurable margins relative to FEC thresholds. Frequency-offset detection experiments further verified the operational monitoring capability required for boundary-node deployment, enabling reliable detection of frequency misconfiguration or abnormal signal injection while preserving cross-domain operational isolation.

These results support the practical feasibility of deploying OAO wavelength converters for multi-domain optical direct-connect boundaries.

Tab. 1: Pre-FEC BER and margin to FEC threshold in the multi-domain ROADM network. Margin (decade) is defined as $\log_{10}(\text{BER}_{th} / \text{BER})$.

Transceiver	FEC scheme	Pre-FEC BER without WC	Pre-FEC BER with WC	Pre-FEC BER threshold	Margin to threshold (decade)
T1	CFEC	$8.8\text{E-}04$	$5.1\text{E-}03$	$1.0\text{E-}02$	0.29
T1	oFEC	$3.6\text{E-}04$	$7.8\text{E-}03$	$2.0\text{E-}02$	0.41
T2	oFEC	$1.2\text{E-}03$	$7.9\text{E-}03$	$2.0\text{E-}02$	0.40

Acknowledgements

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